

SweatBox Gym

Project Report

Executive Summary

SweatBox Gym is a comprehensive fitness management platform designed for a gym located in Sarba, Jounieh, Lebanon. The system connects four key user types - clients, personal trainers, dietitians, and administrators - through a unified ecosystem of mobile and web applications.

Project Overview

Project Type: Multi-platform Fitness Management System

Target Users: Gym clients, Personal trainers, Dietitians, Administrators

Backend: Supabase (PostgreSQL + Edge Functions)

Status: Development/Setup Phase

Applications

1. Mobile App (Flutter)

- Technology: Flutter 3.x (Dart)
- Target Users: Clients and Trainers
- Platform: iOS, Android, Web
- Key Libraries: Riverpod, GoRouter, fl_chart

2. Admin Dashboard (Next.js)

- Technology: Next.js 14, React 18, TypeScript
- Port: localhost:3000
- Target Users: Gym Administrators
- Key Libraries: Tailwind CSS, Recharts, Lucide Icons

3. Dietitian Dashboard (Next.js)

- Technology: Next.js 14, React 18, TypeScript
- Port: localhost:3002
- Target Users: Nutrition Specialists

Core Features

User Management & Authentication

- Registration Approval Workflow - Admin reviews and approves new users
- Role-Based Access - Four roles: client, trainer, admin, dietitian
- Secure Authentication - Password hashing, Edge Functions
- Row Level Security (RLS) - Database-level access control

Client Features (Mobile App)

- Self-registration with approval workflow
- Check-in/check-out at the gym
- Book sessions with personal trainers
- View upcoming sessions and history
- Track body composition metrics
- View assigned diet plans
- Log meals and track nutrition
- Monitor fitness progress

Trainer Features (Mobile App)

- View and manage assigned clients
- Set availability schedules
- Confirm/complete bookings
- Add session notes
- Track client workout progress

Admin Dashboard Features

- User Management - Create, edit, approve/reject users
- Subscription Management - open_gym (\$75/mo) or with_pt (\$200/mo)
- Payment Recording - Cash, card, bank transfer
- Booking Oversight - View all PT session bookings
- Attendance Tracking - Monitor gym check-ins
- Exercise Library - Manage 70+ exercises
- Loyalty Rewards - Free PT month at 12, 24, 36 months
- Analytics & Reports

Dietitian Dashboard Features

- View assigned clients
- Record body composition (InBody-style)
- Create diet plans with macro targets
- Build meal plans with scheduled meals
- Access food database (110+ foods)
- Monitor client meal logs and compliance

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- Track client progress

Database Schema Overview

Core Tables

profiles	Base user information (email, name, role, phone)
registration_requests	Pending user registrations
client_profiles	Client-specific data
trainer_profiles	Trainer data (specializations, certifications)
dietitian_profiles	Dietitian data (license, experience)
subscriptions	Membership plans with PT sessions
bookings	PT session bookings
attendance	Gym check-in/check-out records
loyalty_tracking	Consecutive months and rewards

Nutrition Tables

foods	Food database (110+ items)
diet_plans	Diet plans with macro targets
diet_plan_meals	Scheduled meals within plans
diet_plan_meal_items	Individual food items
meal_logs	Client meal tracking
body_compositions	InBody-style body data

Workout Tables

exercises	Exercise library (70+ exercises)
workout_logs	Session workout records
availability	Trainer availability slots

Edge Functions (Serverless API)

register	Handle new user registration
login	Authenticate users with approval check
approve-registration	Admin approves/rejects registrations
admin-action	Admin-specific operations
dietitian-action	Dietitian-specific operations
body-composition	Body composition CRUD
diet-plan	Diet plan management
meal-log	Client meal logging

Security Model

1. Row Level Security (RLS) - Every table has policies ensuring users access only authorized data
2. Role-Based Access Control - Admins have full access; trainers/dietitians see assigned clients
3. Edge Functions - Sensitive operations handled by secure serverless functions
4. Password Hashing - All passwords are hashed before storage
5. Registration Approval - New users must be approved by admin before accessing the system

Subscription Plans

Plan	Price	Includes
Open Gym	\$75/month	Gym access only
With PT	\$200/month	Gym access + Personal training

Loyalty Rewards Program:

- 12 consecutive months: 1 free PT month
- 24 consecutive months: 1 free PT month
- 36 consecutive months: 1 free PT month

Current Status & Limitations

Implemented Features:

- [OK] Complete multi-app architecture
- [OK] User authentication with approval workflow
- [OK] Body composition tracking (InBody-style)
- [OK] Diet plan management
- [OK] Meal logging
- [OK] Session booking
- [OK] Attendance tracking
- [OK] Loyalty rewards system
- [OK] Subscription management
- [OK] Exercise library

Not Yet Implemented:

- [] Payment Gateway - Manual recording only (no Stripe/PayPal)
- [] Email Service - Notifications in DB only (no SMTP)
- [] Push Notifications - Tokens stored but not active
- [] Third-party Integrations - No fitness trackers

User Roles Summary

Role	Access	Primary Interface
Client	Own data, bookings, diet plans, meals	Mobile App
Trainer	Own schedule, assigned clients, workouts	Mobile App
Dietitian	Assigned clients, diet plans, body comp	Dietitian Dashboard
Admin	Full system access, user management	Admin Dashboard

Summary

SweatBox Gym is a full-featured fitness management platform with:

- 3 applications: Mobile (Flutter), Admin Dashboard (Next.js), Dietitian Dashboard (Next.js)
- 4 user roles: Client, Trainer, Dietitian, Admin
- Robust backend: Supabase with PostgreSQL, Edge Functions, and RLS security
- Complete nutrition system: Diet plans, food database, meal tracking, body composition
- Gym operations: Bookings, attendance, subscriptions, loyalty rewards

The system is designed to streamline gym operations by providing role-specific interfaces for all stakeholders while maintaining strong security through database-level access controls.

Report Generated: February 2026
SweatBox Gym - Sarba, Jounieh, Lebanon